

Alexa-Enabled Smart IoT ELECTRICITY METER SWITCH BOARD

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Imagine tracking power-line voltage, frequency, current draw, and appliance power consumption through a smart switch plug. Real-time data is continuously logged to a database, enabling instant energy-use calculation, billing estimation, and live device control.

The system integrates with popular AI assistants and smart speakers, including Alexa, Google Home, Google Assistant, Apple HomeKit, and the ESP RainMaker app. Appliances can be operated through voice commands, automation rules, or a touch-based switchboard.

ESP RainMaker is an end-user platform built for production-ready IoT devices. It enables firmware development and complete IoT solution design for large-scale deployment. Core features include device monitoring, over-the-air firmware updates, and centralised device management, streamlining the transition from prototype to production and market launch. The platform also manages Wi-Fi provisioning, device onboarding, and firmware control, similar to commercial smart plugs and bulbs. A dedicated home application provides

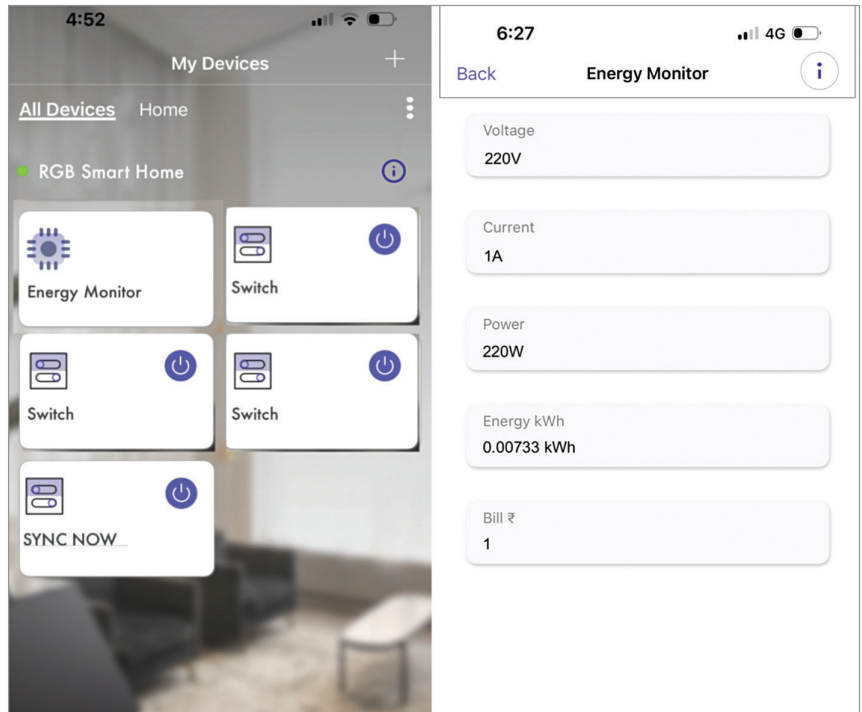


Fig. 1: ESP RainMaker app – home dashboard and real-time energy monitoring



Fig. 2: RainMaker app showing power and energy consumption

BILL OF MATERIALS

Item	Name	Quantity
IndusBoard coin/ESP32 S3 board	U1	1
ACS712 current sensor	U2	1
ZMPT101B voltage sensor	U3	3
Relay module	U4	2
5V DC power adaptor	U5	1
Resistor 2kΩ (R1)	R1	1
Resistor 10kΩ (R2)	R2	1